

Final Project Presentation

Waking up the family



Team 2

Soyeong Yang (20233897)
Eunbyul Park (20234184)
Hyeonho Na (20235529)
Haruta Masuda (20246007)
Keisuke Kudo (20236306)

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I. Project Concept

Overview



“Waking up the family and turn off the alarm!”

Point of view

Third-person game

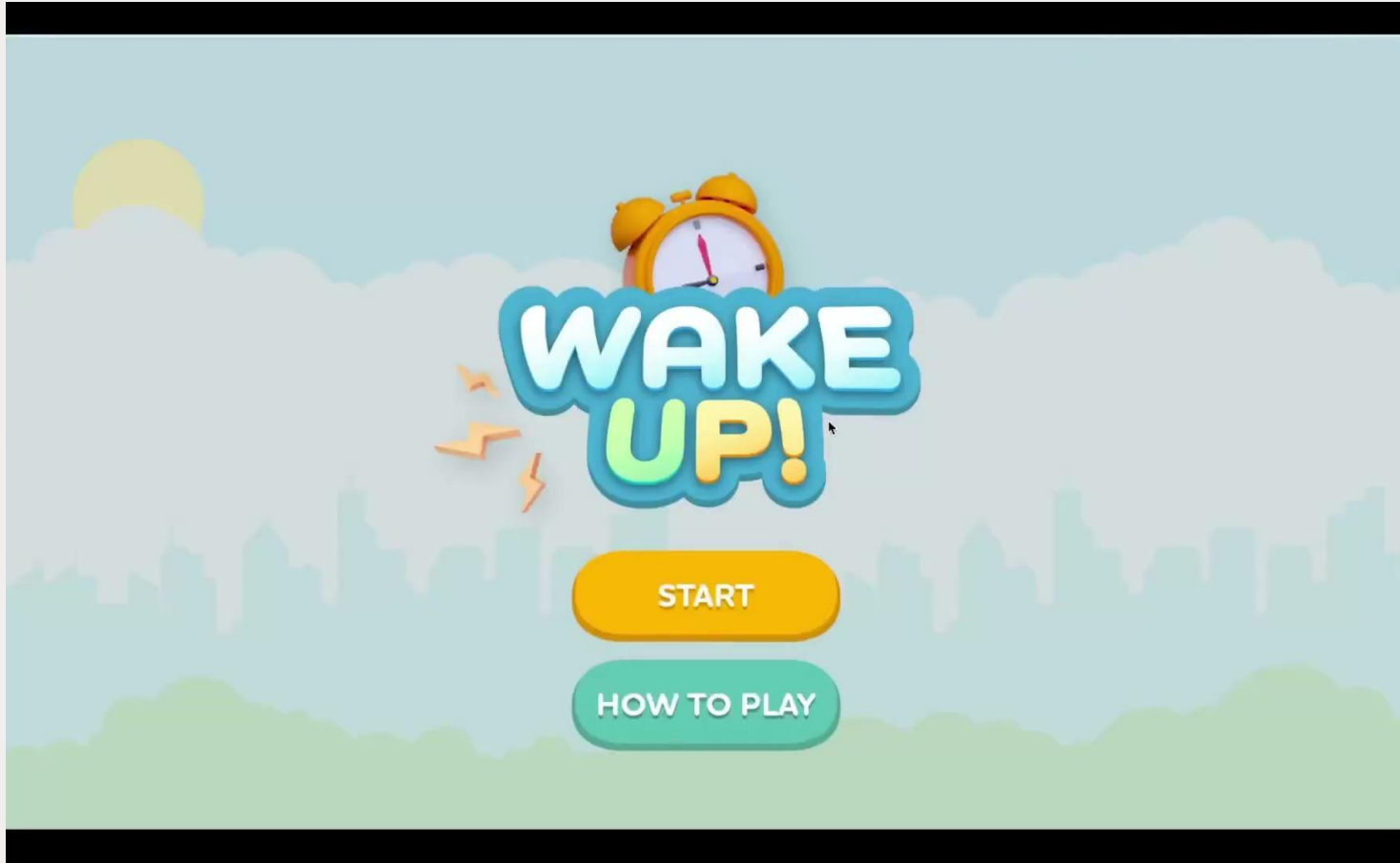
Objectives

Assist family members in waking up on time for their daily routines such as work or school

Tasks

Find family members, and gather their necessary items based on **stereo sound effects**

II. Demo Video



III. Narrative Component

Characters & Items



III. Narrative Component

Characters & Items

Dad (Office Worker)

- Closed Laptop
- Phone
- Burger



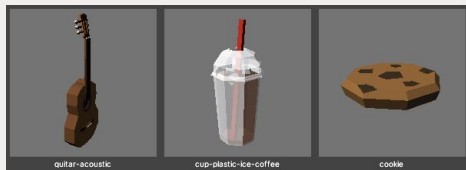
Son (Kid)

- Gaming Console
- VR Headset
- Hotdog



Daughter (Music Student)

- Acoustic Guitar
- Plastic Ice Coffee Cup
- Cookie



III. Narrative Component

Characters & Items

Grandfather (Golfer)

- Golf Club
- Golf Bag
- Toast



Grandmother (Stay-at-home)

- Vacuum Cleaner Robot
- Toilet Paper
- Apple



Uncle (Doctor)

- Pill Box
- Hot Coffee Cup
- Banana



Aunt (Police Officer)

- Red Plastic Cup
- Radio
- Doughnut



III. Narrative Component

Motions

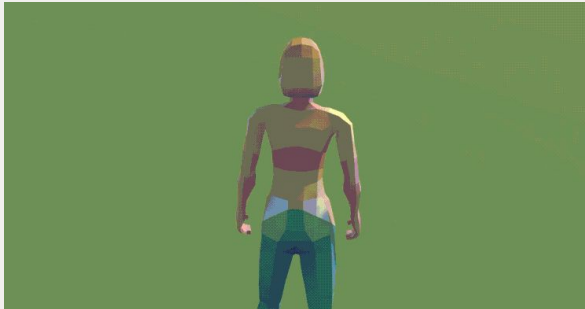
- **Idle**



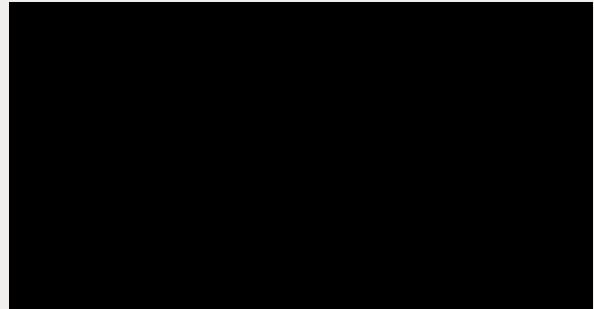
- **Walking**



- **Running**

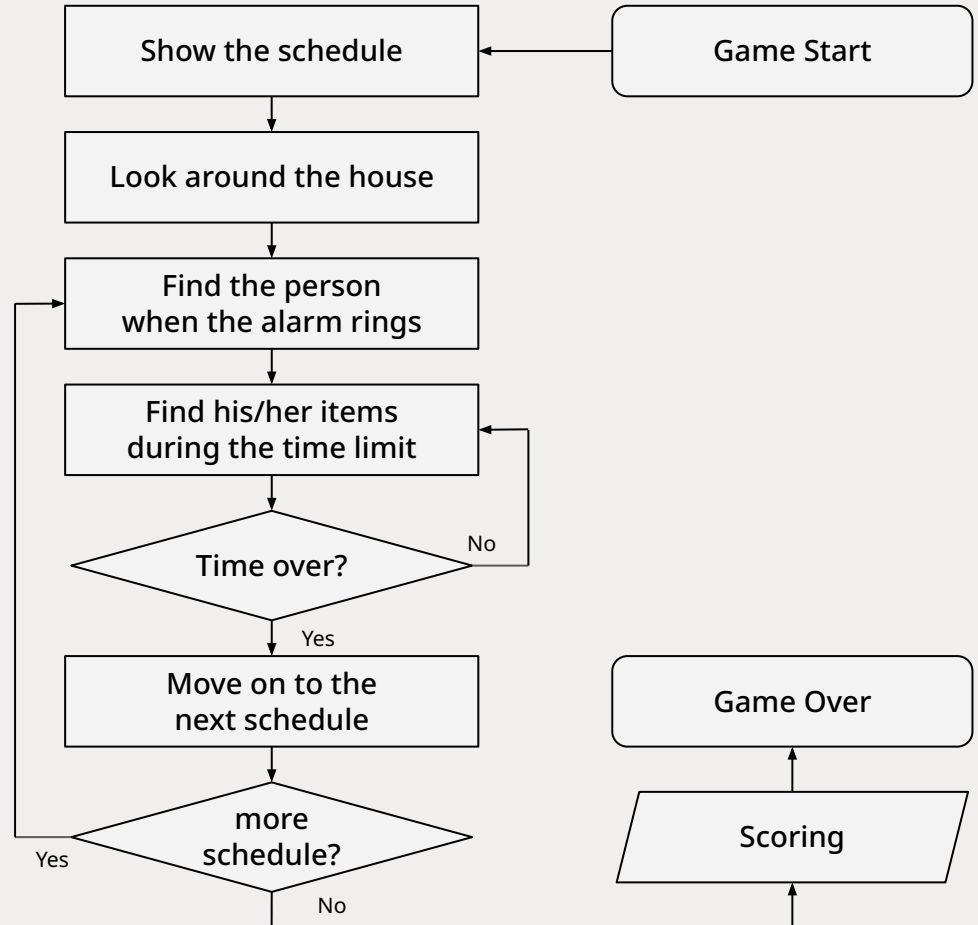
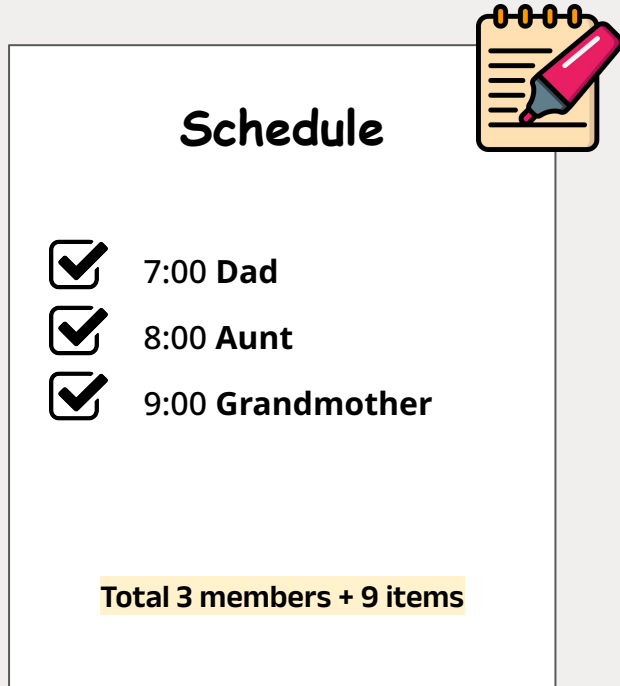


- **Sleeping**



IV. Game Development

Game Sequence



IV. Game Development

Time System



(In-Game) 1 min = (Real world) 1 sec



(In-Game) 1 hour per finding 1 person & 3 items

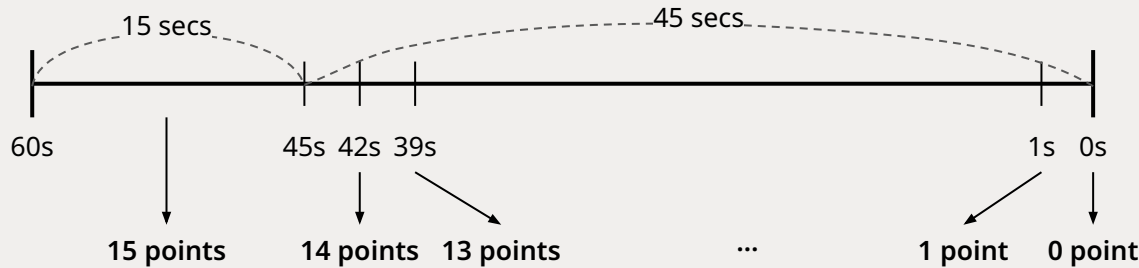


Moving around

IV. Game Development

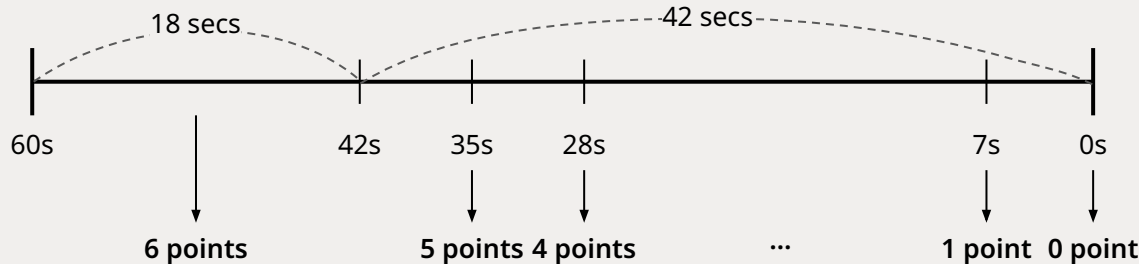
Scoring System

- **Character (15 points)**



$$\text{score} = 15 - (\text{currentTime} / 60 - 15) / 3$$

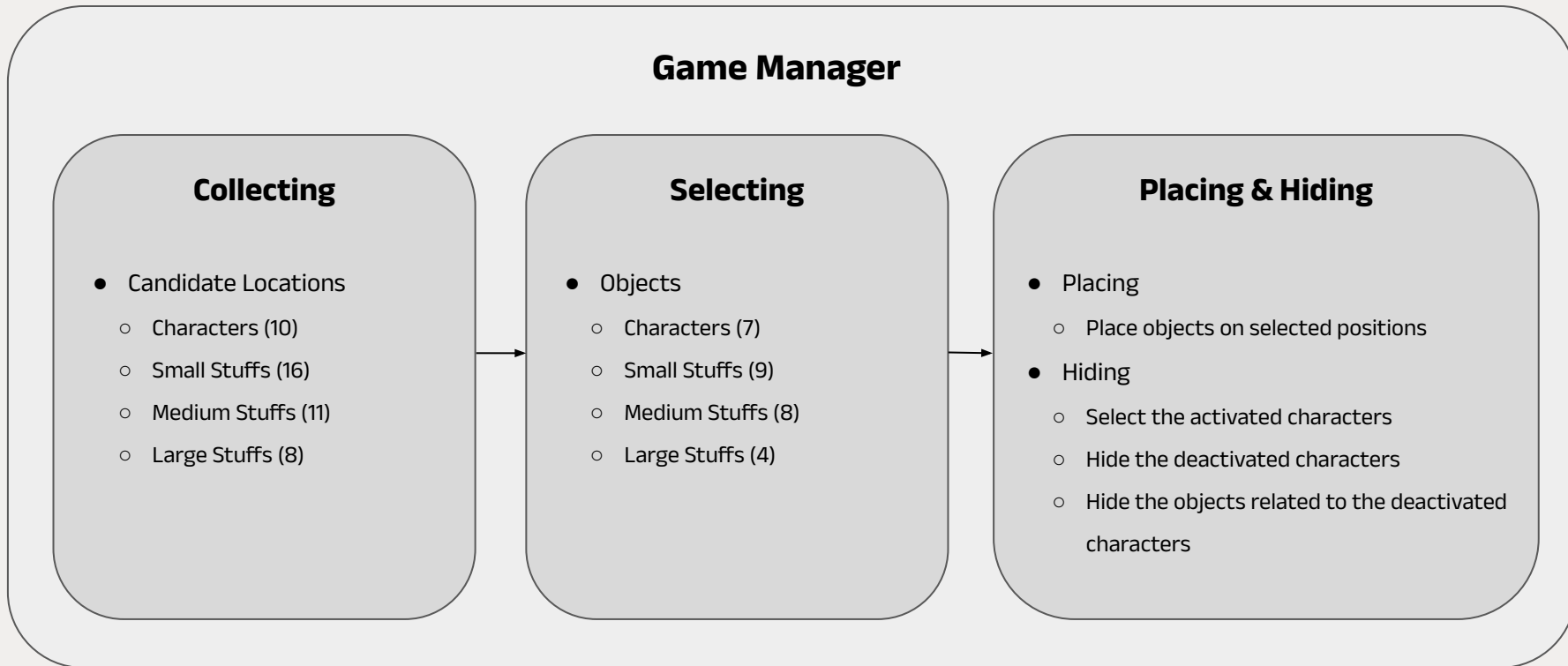
- **Item (6 points)**



$$\text{score} = 6 - (\text{currentTime} / 60 - 18) / 7$$

IV. Game Development

Game positioning system



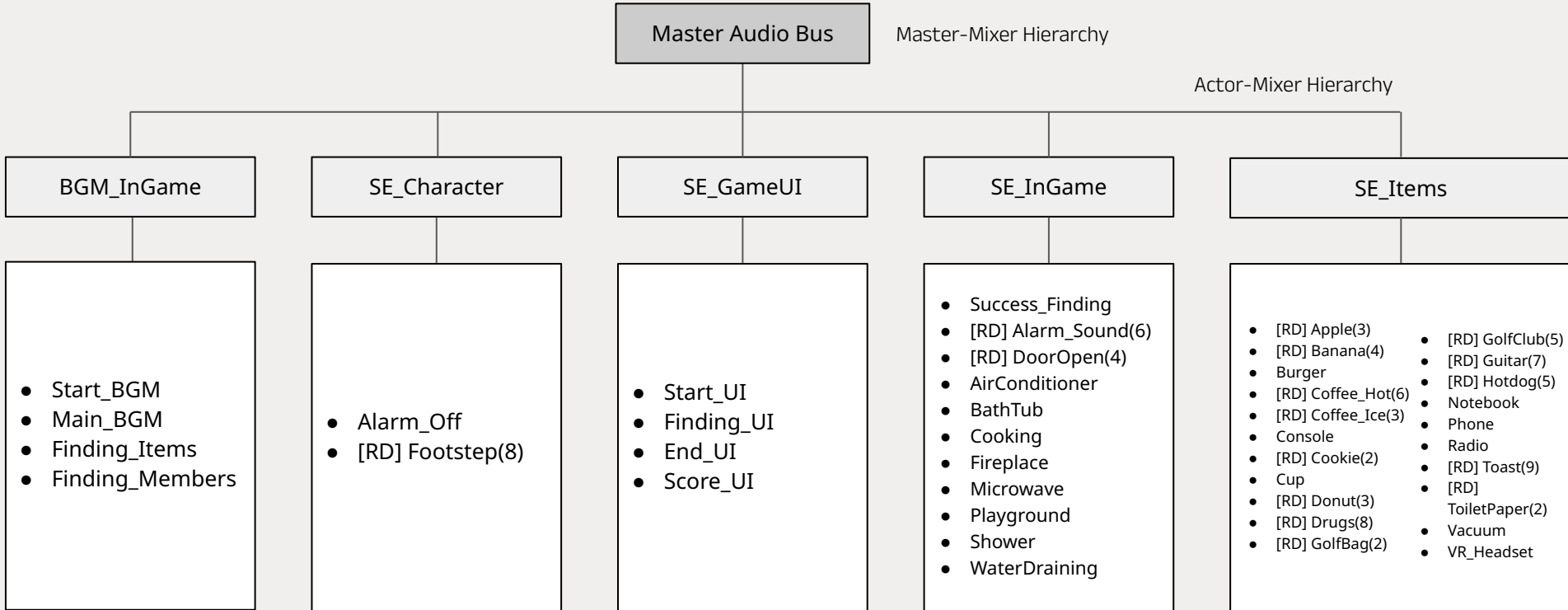
V. Audio System

Overall Sound Components

Sound	Source	Explanation
1) Alarm	Record	Cell phone alarm sound (iPhone, Galaxy)
2) Items	Online	The sounds of randomly placed items to find
3) Ambience	Online	The sound of house ambience
4) Game UI	Online	Sound effects and BGM of UI generated by interaction within the game

V. Audio System

Overall Sound Components



V. Audio System

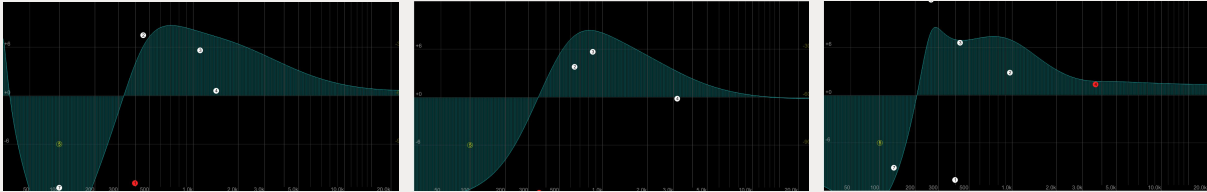
Sound Components (Alarm)

Recording

- Recording Technique - **XY**

Mixing

- REAPER - **EQ** : To remove noise and make it sound as close as possible



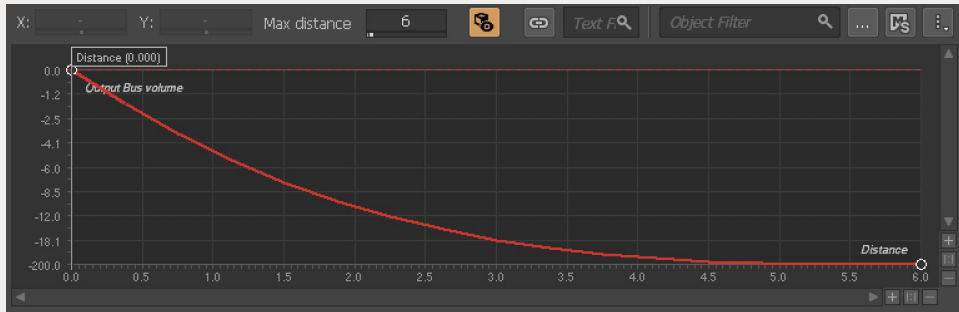
- Wwise - **EQ** : Even though recorded it, it sounded like the original sound effect applied the **Band Pass** like the sound from the phone



V. Audio System

Sound Components (Items)

- **Distance : Log graph** is used to minimize confusion in distinguishing between sounds on the 1st and 2nd floors and to better **identify the exact location of the sound** (but hard to find)



▼ cf) other ambience sound

▲ sound of stuff to find



V. Audio System

Sound Components (Items)

Default Mixing

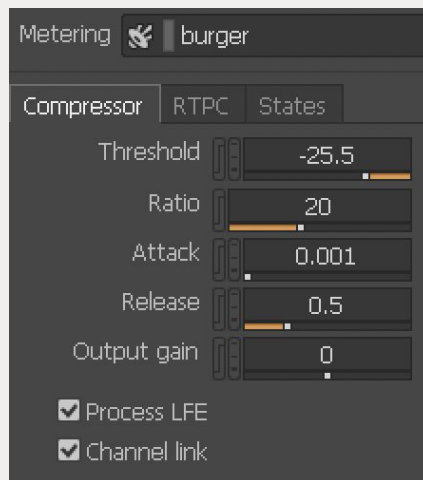
- **Reverb** applied to all items in Wwise, simulating indoor sounds
- Guitar sounds and shouts of "Hotdog" enhanced with additional reverb
- Popping sounds such as wrapping or can opening handled with a **compressor**
- Machine noises : **EQ (Band Pass)** to achieve a mechanical tone
- The doughnut sound : lower the **pitch** and apply the **Low Pass Filter**



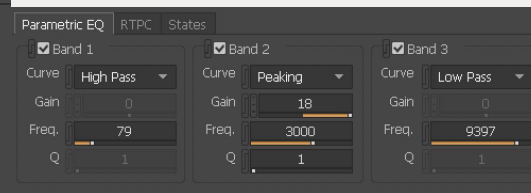
▲ Default Reverb Setting of Items

▲ Reverb Setting of Guitar

▼ Compressor of Burger



▼ EQ of VR headset



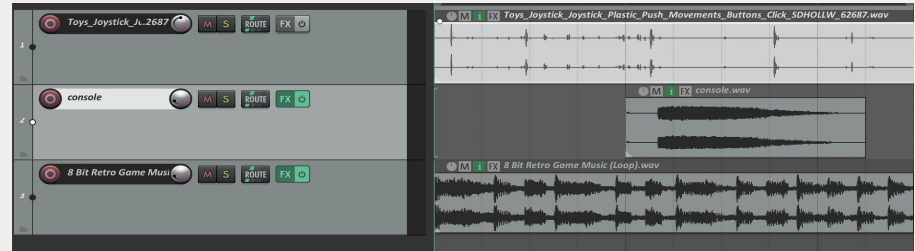
V. Audio System

Sound Components (Items)

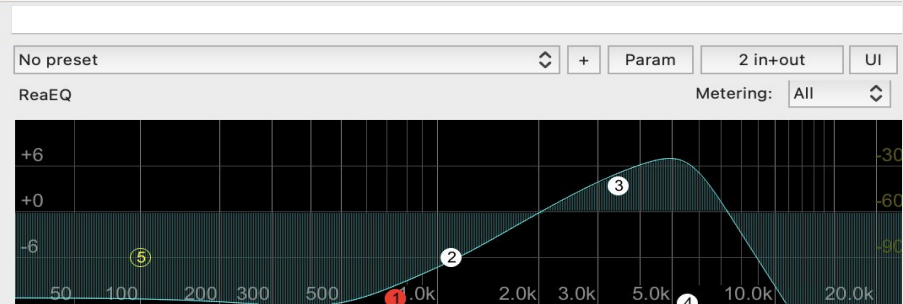
Additional Mixing

- e.g.) Game Console

Sound combination of console key press
+ 8bit game BGM and sound effects



FX: Track 2 "console"

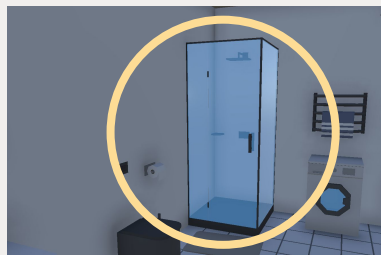
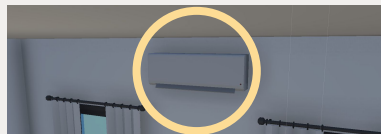


V. Audio System

Sound Components (Ambience)

Sound & Mixing

- The sound of things in the house in order to disturb
 - 1F : Frying(High Pass), Microwave(Low Pass), Draining water(Low Pass), Shower booth (Reverb)
 - 2F : Fireplace (High Pass, Pitch Down), Bathtub (Reverb), Air conditioner (Reverb)
 - Outside : Playground (Reverb)



V. Audio System

Sound Components (Game UI)

Sound & Mixing

- Basic game interaction and success feedback sound

Home	BGM	Creating an interesting atmosphere
	Sound Effects	Feedback from the click-on interaction
In-game	BGM	The creation of a sense of urgency It increases the difficulty of finding things However, adjust it to sound subtle rather than clear (based on wearing the headset)
	Sound Effects	- Instruction to move around freely at first - Instruction to find people and things - Success feedback discovering people and Items - Final completion pop-up feedback

LOOK
AROUND



VI. Limitation

1. Instead of implementing a 3-stage level system (Easy, Medium, Hard), only one stage was implemented under the assumption that it is an Medium Level.
2. Without distinguishing walls, it is based only on distance; therefore, sounds from the first floor are clearly heard on the second floor, and there is an issue where distinctions between rooms are not made.
3. The score is not based on the distance in Wwise but is determined by how quickly one finds the source, with higher scores awarded for faster findings.